Chapter 6. The 2^k Factorial Design

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1 The 2^2 Factorial Design

Motivation. Studying factors one at a time misses interactions and wastes runs. In a factorial design every run informs every effect; with two levels per factor the effects are simple contrasts that can be read off a sign table.

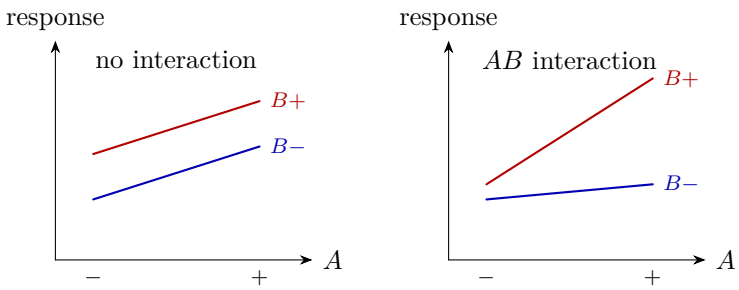


Figure. *Interaction plots for a 2^2 design. Parallel lines mean the effect of A is the same at both levels of B; non-parallel lines mean an AB interaction, and then the main effect of A alone does not describe the data.*

- The 2^k design is a special case of the general factorial: k factors, each at **two levels** ("low" $-$ and "high" $+$; quantitative or qualitative). Very widely used in industry (screening).
- Simplest case: 2^2. Effects model

$$y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + \varepsilon_{ijk}, \qquad i = 1, 2;\ j = 1, 2;\ k = 1, \ldots, n.$$

1.1 Notation for treatment combinations

A treatment combination is denoted by the lowercase letters of the factors at their high level: (1) = both low, $a = A$ high, $b = B$ high, ab = both high. The same symbol also denotes the **total** of the n observations at that combination.

$$\begin{array}{ccc} b\ (-,+) & — & ab\ (+,+) \\ | & & | \\ (1)\ (-,-) & — & a\ (+,-) \end{array}$$

Example 6.1: Chemical process data ($n = 3$)

A = reactant concentration, B = amount of catalyst, response = yield.

A	B	I	II	III	Total
$-$	$-$	28	25	27	$(1) = 80$
$+$	$-$	36	32	32	$a = 100$
$-$	$+$	18	19	23	$b = 60$
$+$	$+$	31	30	29	$ab = 90$

Analysis procedure:

(1) estimate factor effects (main effects A, B, interaction AB);

(2) statistical testing (ANOVA: SS for A, B, AB, error);

(3) residual analysis (normality, constant variance).

1.2 Estimating effects

Derivation

The effect of A at the low level of B is $\dfrac{a-(1)}{n}$; at the high level of B it is $\dfrac{ab-b}{n}$. The main effect of A is their average:

$$A = \frac{1}{2}\Big(\frac{a-(1)}{n} + \frac{ab-b}{n}\Big) = \frac{ab+a-b-(1)}{2n}.$$

Equivalently $A = \bar{y}_{A^+} - \bar{y}_{A^-} = \dfrac{ab+a}{2n} - \dfrac{b+(1)}{2n}$.

$$A = \frac{1}{2n}[ab+a-b-(1)], \qquad B = \frac{1}{2n}[ab+b-a-(1)], \qquad AB = \frac{1}{2n}[ab+(1)-a-b].$$

AB is half the difference between the A effect at high B and at low B: $\frac{1}{2}\big[\frac{ab-b}{n} - \frac{a-(1)}{n}\big]$.

Table of algebraic signs

Treatment combination	I	A	B	AB
(1)	$+$	$-$	$-$	$+$
a	$+$	$+$	$-$	$-$
b	$+$	$-$	$+$	$-$
ab	$+$	$+$	$+$	$+$

The AB column is the elementwise product of the A and B columns. Every pair of columns is orthogonal.

1.3 Sums of squares via contrasts

Theorem 6.2: SS of a contrast

For a contrast $C = \sum_\ell c_\ell S_\ell$ in the treatment totals S_ℓ (each a sum of n observations), $\sum_\ell c_\ell = 0$,

$$SS_C = \frac{(\text{contrast})^2}{n \sum_\ell c_\ell^2}.$$

In the 2^2 design: $\text{Contrast}_A = ab + a - b - (1)$, $\text{Contrast}_B = ab + b - a - (1)$, $\text{Contrast}_{AB} = ab + (1) - a - b$, $\sum c_\ell^2 = 4$, so

$$SS_A = \frac{[ab + a - b - (1)]^2}{4n}, \quad SS_B = \frac{[ab + b - a - (1)]^2}{4n}, \quad SS_{AB} = \frac{[ab + (1) - a - b]^2}{4n}.$$

Direct proof for SS_A [Supplement]. In the two-factor notation $SS_A = \sum_{i=1}^2 \sum_{j=1}^2 \sum_{k=1}^n (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2 = 2n \sum_{i=1}^2 (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2 = 2n[(\bar{y}_{A^+} - \bar{y}_{\cdots})^2 + (\bar{y}_{A^-} - \bar{y}_{\cdots})^2]$, with $\bar{y}_{A^+} = \frac{ab+a}{2n}$, $\bar{y}_{A^-} = \frac{b+(1)}{2n}$, $\bar{y}_{\cdots} = \frac{ab+a+b+(1)}{4n}$:

$$\bar{y}_{A^+} - \bar{y}_{\cdots} = \frac{2(ab+a) - (ab+a+b+(1))}{4n} = \frac{ab + a - b - (1)}{4n}, \qquad \bar{y}_{A^-} - \bar{y}_{\cdots} = -\frac{ab + a - b - (1)}{4n},$$

$$SS_A = 2n \cdot 2 \cdot \frac{[ab + a - b - (1)]^2}{16n^2} = \frac{[ab + a - b - (1)]^2}{4n}.$$

■

General: $t_0^2 = F_0$ [Supplement]. Let $S_\ell = \sum_{k=1}^n y_{\ell k} = n\bar{y}_\ell$ (ℓ indexes the cells), and $C = \sum_\ell c_\ell S_\ell = n \sum_\ell c_\ell \bar{y}_\ell$. Then

$$\text{Var}(C) = \text{Var}\Big(n \sum_\ell c_\ell \bar{y}_\ell\Big) = n^2 \sum_\ell c_\ell^2 \text{Var}(\bar{y}_\ell) = n^2 \sum_\ell c_\ell^2 \frac{\sigma^2}{n} = n\sigma^2 \sum_\ell c_\ell^2.$$

Let $t_0 = \dfrac{C - \text{E}(C)}{\sqrt{\widehat{\text{Var}}(C)}} = \dfrac{\sum c_\ell S_\ell - \text{E}(C)}{\sqrt{n\, MS_E \sum c_\ell^2}}$. Assuming $\text{E}(C) = 0$ (true under H_0, see below),

$$t_0^2 = \frac{(\sum_\ell c_\ell S_\ell)^2 / (n \sum c_\ell^2)}{MS_E} = \frac{SS_C/1}{MS_E} \sim F_{1,\, 2^2(n-1)}.$$

So each effect's F-test in the ANOVA is the square of a t-test of the contrast. ■

Derivation: $\text{E}(\text{Contrast}_A) = 0$ under H_0

With effects A_1, A_2 (low/high) etc.,

$$ab = \textstyle\sum_k y_{22k} = n(\mu + A_2 + B_2 + (AB)_{22}) + \varepsilon_{22\cdot}, \qquad a = n(\mu + A_2 + B_1 + (AB)_{21}) + \varepsilon_{21\cdot},$$
$$b = n(\mu + A_1 + B_2 + (AB)_{12}) + \varepsilon_{12\cdot}, \qquad (1) = n(\mu + A_1 + B_1 + (AB)_{11}) + \varepsilon_{11\cdot}.$$

Hence

$$\text{E}(\text{Contrast}_A) = n(2A_2 - 2A_1 + \underbrace{(AB)_{22} + (AB)_{21} - (AB)_{12} - (AB)_{11}}_{=0 \text{ by } \sum_i (AB)_{ij} = 0}) = n(2A_2 - 2A_1) \overset{H_0}{=} 0.$$

Since $A_1 + A_2 = 0$, $\text{E}(\text{Contrast}_A) = 4nA_2$ and $\text{E}(A \text{ effect}) = \frac{4nA_2}{2n} = 2A_2 = A_2 - A_1$.

1.4 ANOVA

DF: $2^2 n - 1 = 1 + 1 + 1 + 2^2(n-1)$.

Source	SS	DF	MS	F_0
A	SS_A	1	MS_A	MS_A/MS_E
B	SS_B	1	MS_B	MS_B/MS_E
AB	SS_{AB}	1	MS_{AB}	MS_{AB}/MS_E
Error	SS_E	$2^2(n-1)$	MS_E	
Total	SS_T	$2^2 n - 1$		

Example 6.3: Chemical process, line by line

$$A = \frac{90 + 100 - 60 - 80}{2(3)} = \frac{50}{6} = 8.33,$$
$$B = \frac{90 + 60 - 100 - 80}{6} = \frac{-30}{6} = -5.00,$$
$$AB = \frac{90 + 80 - 100 - 60}{6} = \frac{10}{6} = 1.67,$$
$$SS_A = \frac{50^2}{4(3)} = 208.33, \quad SS_B = \frac{(-30)^2}{12} = 75.00, \quad SS_{AB} = \frac{10^2}{12} = 8.33,$$
$$SS_T = \sum y^2 - \frac{y_{...}^2}{12} = 9398 - \frac{330^2}{12} = 9398 - 9075 = 323.00,$$
$$SS_E = 323.00 - 208.33 - 75.00 - 8.33 = 31.34.$$

Source	SS	DF	MS	F_0
A	208.33	1	208.33	53.15
B	75.00	1	75.00	19.13
AB	8.33	1	8.33	2.13
Error	31.34	8	3.92	
Total	323.00	11		

$F_{0.05,1,8} = 5.32$: A and B significant, AB not. Increasing concentration raises yield; increasing catalyst lowers it.

2 The 2^3 Design

Motivation. Adding a third factor shows the general pattern: 2^k runs estimate k main effects and all interactions, each as an orthogonal contrast with the same variance.

Factors A, B, C at two levels: 8 runs $(1), a, b, ab, c, ac, bc, abc$.

Trt	I	A	B	AB	C	AC	BC	ABC
(1)	+	−	−	+	−	+	+	−
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
ab	+	+	+	+	−	−	−	−
c	+	−	−	+	+	−	−	+
ac	+	+	−	−	+	+	−	−
bc	+	−	+	−	+	−	+	−
abc	+	+	+	+	+	+	+	+

Properties:

(1) except for I, every column has equally many + and − (a contrast);

(2) the product of any two columns is a column of the table ($A \cdot B = AB$, $AB \cdot BC = AB^2C = AC$, with squares $= I$);

(3) I times any column leaves it unchanged;

(4) all columns are mutually orthogonal.

$$A = \frac{1}{4n}\left[(abc + ac + ab + a) - (bc + c + b + (1))\right], \qquad SS_A = \frac{(\text{Contrast}_A)^2}{8n}.$$

Theorem 6.4: General 2^k

$$\text{Effect} = \frac{\text{Contrast}}{2^{k-1}n}, \qquad SS = \frac{(\text{Contrast})^2}{2^k n}, \qquad \text{Var(Effect)} = \frac{2^k n\sigma^2}{(2^{k-1}n)^2} = \frac{\sigma^2}{2^{k-2}n},$$

$\text{df(Total)} = n2^k - 1$, $\text{df(Error)} = 2^k(n-1)$, each of the $2^k - 1$ effects has 1 df.

Proof [Supplement]. Each contrast uses all 2^k totals with $c_\ell = \pm 1$, so $\sum c_\ell^2 = 2^k$ and $SS = \text{Contrast}^2/(n2^k)$. The effect is (sum of 2^{k-1} totals at +)/$(2^{k-1}n)$ minus (sum at −)/$(2^{k-1}n)$, i.e. $\text{Contrast}/(2^{k-1}n)$. Variance: $\text{Var(Contrast)} = n\sigma^2 \sum c_\ell^2 = 2^k n\sigma^2$. ■

Source	SS	DF
A, B, C	SS_A, SS_B, SS_C	1 each
AB, AC, BC	$SS_{AB}, SS_{AC}, SS_{BC}$	1 each
ABC	SS_{ABC}	1
Error	SS_E	$2^3(n-1)$
Total	SS_T	$n2^3 - 1$

Example 6.5: A 2^3 with $n = 2$ (soft-drink fill height)

Totals: $(1) = -4$, $a = 1$, $b = -1$, $ab = 5$, $c = -1$, $ac = 3$, $bc = 2$, $abc = 11$ (sum $= 16$).

$$\begin{aligned}
\text{Contrast}_A &= a + ab + ac + abc - (1) - b - c - bc = 1 + 5 + 3 + 11 + 4 + 1 + 1 - 2 = 24,\\
\text{Contrast}_B &= b + ab + bc + abc - (1) - a - c - ac = -1 + 5 + 2 + 11 + 4 - 1 + 1 - 3 = 18,\\
\text{Contrast}_C &= c + ac + bc + abc - (1) - a - b - ab = -1 + 3 + 2 + 11 + 4 - 1 + 1 - 5 = 14,\\
\text{Contrast}_{AB} &= (1) + ab + c + abc - a - b - ac - bc = -4 + 5 - 1 + 11 - 1 + 1 - 3 - 2 = 6,\\
\text{Contrast}_{AC} &= (1) + b + ac + abc - a - ab - c - bc = -4 - 1 + 3 + 11 - 1 - 5 + 1 - 2 = 2,\\
\text{Contrast}_{BC} &= (1) + a + bc + abc - b - ab - c - ac = -4 + 1 + 2 + 11 + 1 - 5 + 1 - 3 = 4,
\end{aligned}$$

$\text{Contrast}_{ABC} = a + b + c + abc - (1) - ab - ac - bc = 1 - 1 - 1 + 11 + 4 - 5 - 3 - 2 = 4.$

Effects = Contrast$/(4 \cdot 2)$: $A = 3.00$, $B = 2.25$, $C = 1.75$, $AB = 0.75$, $AC = 0.25$, $BC = 0.50$, $ABC = 0.50$. SS = Contrast$^2/16$: 36.00, 20.25, 12.25, 2.25, 0.25, 1.00, 1.00. With $SS_T = 78.00$, $SS_E = 78 - 73 = 5.00$ on 8 df, $MS_E = 0.625$; $F_A = 57.6$, $F_B = 32.4$, $F_C = 19.6$, $F_{AB} = 3.6$ ($F_{0.05,1,8} = 5.32$): only the main effects are significant.

3 Regression Approach to the 2^k Design

Derivation

2^2: $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_{12} x_{1i} x_{2i} + \varepsilon_i$ with coded $x = \pm 1$. Using the totals as responses,

$$\mathbf{X} = \begin{pmatrix} 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} (1) \\ a \\ b \\ ab \end{pmatrix}$$

$$\Rightarrow \mathbf{X}^T\mathbf{y} = \begin{pmatrix} (1) + a + b + ab \\ -(1) + a - b + ab \\ -(1) - a + b + ab \\ (1) - a - b + ab \end{pmatrix} = \begin{pmatrix} \text{Contrast}_{\text{total}} \\ \text{Contrast}_A \\ \text{Contrast}_B \\ \text{Contrast}_{AB} \end{pmatrix}.$$

Since $\mathbf{X}^T\mathbf{X} = 4\mathbf{I}$, $\hat{\boldsymbol{\beta}} = \frac{1}{4}\mathbf{X}^T\mathbf{y}$, e.g.

$$\hat{\beta}_1 = \tfrac{1}{4}\text{Contrast}_A = \tfrac{1}{4}(2n \cdot \text{main effect } A) = \tfrac{n}{2} \cdot \text{main effect } A.$$

2^3: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \beta_3 x_3 + \beta_{13} x_1 x_3 + \beta_{23} x_2 x_3 + \beta_{123} x_1 x_2 x_3 + \varepsilon$, $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \frac{1}{2^3}(\text{Contrast}_{\text{total}}, \text{Contrast}_A, \dots)^T$ and since $A = \frac{1}{2^{k-1}n}\text{Contrast}_A$, $\hat{\beta}_1 = \frac{n}{2}\,\text{main effect}_A$.

Remark 6.6: Using cell averages instead

If the responses are the cell *averages* $\bar{y}_\ell = S_\ell/n$, then $\hat{\beta}_1 = \frac{1}{4}\sum_\ell c_\ell \bar{y}_\ell = \frac{1}{4n}\text{Contrast}_A = \frac{A}{2}$: a regression coefficient is **half the effect**, because x moves from -1 to $+1$ (two units). Chemical data: $\mathbf{y} = (80/3, 100/3, 60/3, 90/3)^T$ gives $\hat{\beta}_0 = 27.5$, $\hat{\beta}_1 = 4.1667 = 8.33/2$, $\hat{\beta}_2 = -2.5 = -5/2$, $\hat{\beta}_{12} = 0.8333 = 1.67/2$ (matching the R output).

4 Residual Analysis

Fitted values from the regression model (only significant terms): chemical process $\hat{y} = 27.5 + \frac{8.33}{2}x_1 - \frac{5.00}{2}x_2$. Check normality (Q–Q plot of residuals) and constant variance (residuals vs fitted).

5 Summary of Designs So Far

Design	Model	Analysis
CRD	$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$	one-way ANOVA / regression
RCBD, BIBD	$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$	two-way ANOVA / regression
Latin square	$y_{ijk} = \mu + \alpha_i + \tau_j + \beta_k + \varepsilon_{ijk}$	three-way ANOVA
Graeco-Latin	$y_{ijkl} = \mu + \theta_i + \tau_j + \omega_k + \Psi_l + \varepsilon_{ijkl}$	four-way ANOVA
Factorial	$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$	two-way ANOVA with interaction
2^k	contrasts of $(1), a, b, ab, \ldots$	ANOVA via contrasts / regression

6 Selected exercises

Motivation. The exercises prove the sign-table properties used throughout, explain why an unreplicated 2^k is judged by a normal plot of effects, add center points to detect curvature, and show how the precision of a fitted first-order model depends on the location.

Exercise 6.1 (Sign table and the unreplicated 2^k)

Code each factor as $x_j = \pm 1$ and let the column of the effect with letter set $J \subseteq \{1, \ldots, k\}$ be $c_J(\mathbf{x}) = \prod_{j \in J} x_j$ over the 2^k runs ($c_\emptyset = I$).
(a) Prove properties (1)–(4) of the sign table: $\sum_{\mathbf{x}} c_J(\mathbf{x}) = 0$ for $J \neq \emptyset$, $c_J c_K = c_{J \triangle K}$, and $\sum_{\mathbf{x}} c_J(\mathbf{x}) c_K(\mathbf{x}) = 2^k \mathbf{1}\{J = K\}$.
(b) In a single replicate ($n = 1$) with $y(\mathbf{x}) = \mu(\mathbf{x}) + \varepsilon(\mathbf{x})$, $\varepsilon \overset{\text{iid}}{\sim} N(0, \sigma^2)$, show that the $2^k - 1$ effect estimates are independent $N(\text{effect}, \sigma^2/2^{k-2})$, and that their sums of squares add up to SS_T.

Solution [Supplement].

(a)

Step 1. For $J \neq \emptyset$, summing over the 2^k points factor by factor,

$$\sum_{\mathbf{x} \in \{\pm 1\}^k} \prod_{j \in J} x_j = \prod_{j \in J} \Big(\sum_{x_j = \pm 1} x_j \Big) \cdot 2^{k - |J|} = 0.$$

Step 2. $c_J(\mathbf{x}) c_K(\mathbf{x}) = \prod_{j \in J \cap K} x_j^2 \prod_{j \in J \triangle K} x_j = c_{J \triangle K}(\mathbf{x})$ since $x_j^2 = 1$ (e.g. $AB \cdot BC = AC$), and $c_\emptyset c_J = c_J$.

Step 3. Hence $\sum_{\mathbf{x}} c_J c_K = \sum_{\mathbf{x}} c_{J \triangle K}$, which is 2^k if $J = K$ and 0 otherwise by Step 1.

(b)

Step 1. The estimate of effect J is $\hat{E}_J = \mathbf{c}_J^T \mathbf{y} / 2^{k-1}$, where $\mathbf{c}_J$ is the column. Being linear in $\mathbf{y} \sim N(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, the estimates are jointly normal with $\text{Cov}(\hat{E}_J, \hat{E}_K) = \sigma^2 \mathbf{c}_J^T \mathbf{c}_K / 2^{2k-2}$, which is 0 for $J \neq K$ and $\sigma^2/2^{k-2}$ for $J = K$ by (a). Uncorrelated jointly normal variables are independent.

Step 2. By (a) the 2^k columns $\mathbf{c}_J / 2^{k/2}$ form an orthonormal basis of $\mathbb{R}^{2^k}$, so $\mathbf{y}^T \mathbf{y} = \sum_J (\mathbf{c}_J^T \mathbf{y})^2 / 2^k$. The $J = \emptyset$ term is $(\sum y)^2 / 2^k$, and for $J \neq \emptyset$ the term is $SS_J = \text{Contrast}_J^2 / 2^k$. Hence $\sum_{J \neq \emptyset} SS_J = \mathbf{y}^T \mathbf{y} - (\sum y)^2 / 2^k = SS_T$.

With all $2^k - 1$ degrees of freedom used by effects there is no error estimate. But by Step 1 the estimates of null effects form a sample from $N(0, \sigma^2/2^{k-2})$ and fall on a line in a normal probability

plot, while real effects stand out; alternatively the negligible high-order interactions are pooled into error. ■

Exercise 6.2 (Center points and curvature)

A 2^k design ($n_F = 2^k$ runs) is augmented with n_C runs at the center $\mathbf{x} = \mathbf{0}$. Suppose the true mean is the second-order model $\mu(\mathbf{x}) = \beta_0 + \sum_j \beta_j x_j + \sum_{i<j} \beta_{ij} x_i x_j + \sum_j \beta_{jj} x_j^2$ and errors are iid $N(0, \sigma^2)$.
(a) Show $\mathrm{E}(\bar{y}_F - \bar{y}_C) = \sum_j \beta_{jj}$.
(b) Show that under $H_0 : \sum_j \beta_{jj} = 0$,

$$F_0 = \frac{SS_{\text{pure quad}}}{s_C^2} \sim F_{1,\, n_C - 1}, \qquad SS_{\text{pure quad}} = \frac{n_F n_C (\bar{y}_F - \bar{y}_C)^2}{n_F + n_C},$$

where s_C^2 is the sample variance of the center runs.
(c) Show that adding center runs changes none of the factorial effect estimates.

Solution [Supplement].

(a) Over the factorial points $x_j^2 = 1$, while $\sum x_j = 0$ and $\sum x_i x_j = 0$ $(i \neq j)$ by Exercise 6.1(a). Hence $\mathrm{E}\bar{y}_F = \beta_0 + \sum_j \beta_{jj}$. At the center $\mu(\mathbf{0}) = \beta_0$, so $\mathrm{E}\bar{y}_C = \beta_0$.

(b)

Step 1. $\bar{y}_F$ and $\bar{y}_C$ are independent with variances σ^2/n_F and σ^2/n_C, so $\mathrm{Var}(\bar{y}_F - \bar{y}_C) = \sigma^2(n_F + n_C)/(n_F n_C)$ and, under H_0, $SS_{\text{pure quad}}/\sigma^2 = (\bar{y}_F - \bar{y}_C)^2/\mathrm{Var}(\bar{y}_F - \bar{y}_C) \sim \chi_1^2$.

Step 2. The center runs are an iid normal sample, so $(n_C - 1)s_C^2/\sigma^2 \sim \chi_{n_C-1}^2$ independently of $\bar{y}_C$; it is also independent of the factorial runs. Hence s_C^2 is independent of $\bar{y}_F - \bar{y}_C$ and $F_0 \sim F_{1,n_C-1}$.

Step 3. A significant F_0 signals curvature but cannot say which β_{jj} is nonzero; that needs a response surface design with more than two levels per factor.

(c) Each effect estimate is a contrast over the factorial runs only (center runs have $x_j = 0$ and contribute 0 to every column c_J, $J \neq \emptyset$). So the centers buy a pure-error estimate and a curvature check without disturbing the factorial analysis. ■

Exercise 6.3 (Precision of the fitted first-order model)

The first-order model $y = \beta_0 + \sum_{j=1}^k \beta_j x_j + \varepsilon$ is fitted to n replicates of a 2^k design ($N = n2^k$ runs). Show that the variance of the fitted value at a point $\mathbf{x}$ of the design region is

$$\mathrm{Var}\{\hat{y}(\mathbf{x})\} = \frac{\sigma^2}{N}(1 + \|\mathbf{x}\|^2),$$

so it depends only on the distance from the center.

Solution [Supplement].

Step 1. The columns of $\mathbf{X}$ are $I, x_1, \ldots, x_k$, each repeated over the n replicates. By Exercise 6.1(a) they are orthogonal, and every entry is ± 1, so $\mathbf{X}^T\mathbf{X} = N\mathbf{I}_{k+1}$.

Step 2. $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1} = (\sigma^2/N)\mathbf{I}_{k+1}$: the coefficient estimates are uncorrelated with equal variance.

Step 3. $\hat{y}(\mathbf{x}) = (1, \mathbf{x}^T)\hat{\boldsymbol{\beta}}$, so $\mathrm{Var}\{\hat{y}(\mathbf{x})\} = (\sigma^2/N)(1, \mathbf{x}^T)(1, \mathbf{x}^T)^T = (\sigma^2/N)(1 + \|\mathbf{x}\|^2)$.

The variance is smallest at the center (σ^2/N) and largest at the corners $(\sigma^2(1 + k)/N)$; a design with this property (equal precision on spheres) is called rotatable. ■